

Whitehead torsion and the s-cobordism theorem

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Abstract

This are notes that I used to prepare a talk about whitehead torsion and the s-cobordism for the EAST 2026.

1 Simple homotopy theory

Let (K, L) be a finite CW-pair, that is, a CW-complex K that has finitely many cells, and $L \subseteq K$ a subcomplex.

Definition 1. We say that K collapses to L by an elementary collapse $K \searrow^e L$ (or that L expands to K by an elementary expansion) if you can write $K = L \cup e^{n-1} \cup e^n$, e^{n-1} and e^n being cells which are not in L , in such a way that, if you write $\partial D^n = D_+^{n-1} \cup_{S^{n-2}} D_-^{n-1}$ then the characteristic map $\varphi : D^n \rightarrow K$ of e^n is such that

- $\varphi|_{D_+^{n-1}} \rightarrow K$ is the characteristic map of e^{n-1} , and
- $\varphi(D_-^{n-1}) \subseteq L$.

And in this case, K is the mapping cone of $\varphi_- : D_-^{n-1} \rightarrow L$, which in turns means that L is a deformation retract of K and we have maps:

$$\text{a deformation retract: } K \rightarrow L,$$

and

$$\text{a inclusion : } L \rightarrow K.$$

This operation is the building block of simple homotopies.

Definition 2. Let (K, L) be a finite CW-pair.

1. We say that K collapses to L , or L expands to K , if there is a sequence of elementary collapses

$$L = K_0 \searrow^e K_1 \searrow^e \cdots \searrow^e K_{n-1} \searrow^e K_n = K.$$

In this case we still have a deformation retract $K \rightarrow L$ and inclusion $L \rightarrow K$;

2. A map $f : K \rightarrow L$ is a simple homotopy equivalence if it is a composition

$$K = K_0 \rightarrow K_1 \rightarrow \cdots \rightarrow K_{n-1} \rightarrow K_n = L,$$

in which each $K_i \rightarrow K_{i+1}$ is either a deformation retract of a collapse, the inclusion map of an expansion, or a cellular homeomorphism.

Remark 1. Every simple homotopy equivalence is a homotopy equivalence.

2 Chain Complexes

Let X be a CW-complex, and $\pi = \pi_1(X)$. The universal cover of X , $\tilde{X}$, has a free π -action, and so $C_*(\tilde{X})$ is a finite chain complex of free $\mathbb{Z}[\pi]$ -modules, in which the generators are the cells of X . The theory we will build aims to deal with this type of chain complex, but it can also be applied in more generality but one must keep in mind $C_*(\tilde{X})$. For instance, If $\mathbb{Z}[\pi]^n \cong \mathbb{Z}[\pi]^m$, then $n = m$. This needs not be the case for a general ring R , but we will assume that our rings satisfies this property.

Definition 3. Let R be a ring, and $GL_n(R)$ be the group of invertible $n \times n$ matrices over R under multiplication. We have inclusion maps

$$GL_n(R) \rightarrow GL_{n+1}(R)$$

$$A \mapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

Let

$$GL(R) = \lim_n GL_n(R).$$

In this group we can take the subgroup $E(R)$ generated by all matrices of the type

$$I + rE_{ij},$$

where $i \neq j$ and E_{ij} has 1 in position ij and 0 elsewhere. $(I + rE_{ij})A$ is the matrix whose j -th column had ben replaced by the sum of the j -th column of A with r times the i -th column of A . If the multiplication was on the left, i.e. $A(I + rE_{ij})$, then the operation is made on the rows i and j instead.

This subgroup is precisely the commutator of G .

Lemma 2.1 (Whitehead Lemma).

$$E(R) = [GL(R), GL(R)]$$

So, we can define

Definition 4. Let

$$K_1(R) = GL(R)/E(R).$$

$K_1(R)$ is an abelian group and the projection

$$GL(R) \rightarrow K_1(R)$$

$$A \mapsto [A],$$

can be thought of as a generalized determinant.

We can reduce this group but moding it out by the sign, that is

$$\tilde{K}_1(R) = K_1(R)/[(-1)].$$

3 $K_1(R)$ and $\text{Wh}(\pi)$

Going back to our case of interest, we will take $R = \mathbb{Z}[\pi]$, where $\pi = \pi_1(X)$. As stated before, $C_*(\tilde{X})$ is a free $\mathbb{Z}[\pi]$ -chain complex, but to get the cells of X as generators, one must also choose a lift for each cell and a orientation for it. That means that the generators are determined up to a multiple of $\pm\alpha, \alpha \in \pi$. This is why we define the Whitehead group as

Definition 5. The Whitehead group of π , denoted $\text{Wh}(\pi)$, is the quotient

$$\text{Wh}(\pi) := \tilde{K}_1(\mathbb{Z}[\pi]) / \pm \pi.$$

If X is path-connected then $\text{Wh}(\pi_1(X))$ does not depend on the basepoint.

Lemma 3.1. *Let $\pi = \pi_1(X)$, for some topological space X . If $f : \pi \rightarrow \pi$ is a inner automorphism by conjugation, i.e.*

$$f(\alpha) = x\alpha x^{-1},$$

for some $x \in \pi$, then the induced map

$$f_* : Wh(\pi) \rightarrow Wh(\pi),$$

is the identity.

4 Torsion of chain complexes

Let X, Y be a connected finite CW-complex, and let $\pi = \pi_1(X)$. If $f : X \rightarrow Y$ is a homotopy equivalence (and in particular, $\pi_1(Y) = \pi$), then f is homotopic to a cellular map $g : X \rightarrow Y$. The map g , in turn defines a weak homotopy equivalence of based $\mathbb{Z}[\pi]$ -chain complexes

$$g_* : C_*(X) \rightarrow C_*(Y).$$

We want to define a map

$$f \mapsto \tau(f) \in Wh(\pi),$$

in such a way that f is a simple homotopy equivalence if and only if $\tau(f) = 0$. To do so, we define τ for all weak homotopy equivalences of based chain complexes.

The first step is to consider matrices, since that is how K_1 is defined.

Definition 6. Let $f : M \rightarrow M'$ be an isomorphism of R -modules, with $\{b_1, \dots, b_n\}$ and $\{b'_1, \dots, b'_n\}$ basis of M and M' , respectively. We can write

$$f(b_i) = \sum_j b'_j f_{ij},$$

and then we have a matrix (f_{ij}) , allowing us to define a homomorphism

$$\begin{aligned} \text{Iso}_R(M, M') &\rightarrow GL(R) \rightarrow \tilde{K}_1(R) \\ f &\mapsto (f_{ij}) \mapsto [f] = [(f_{ij})], \end{aligned}$$

where we have implicitly ordered the bases to write down the matrix (f_{ij}) ; passing to $\tilde{K}_1(R)$ is exactly what makes $[f]$ independent of this choice of order (and of signs), i.e. of an *unordered* basis.

If we replace M, M' by based R -chain complexes C, C' , we can define the torsion of a map $f : C \rightarrow C'$ as

$$\tau(f) = \sum (-1)^n [f_n] \in \tilde{K}_1(R).$$

Alternatively, we can consider the map $f : M \rightarrow M'$ as an acyclic based chain complex

$$\cdots \rightarrow 0 \rightarrow M \xrightarrow{f} M' \rightarrow 0 \rightarrow \cdots,$$

and then we will have $[f] = \pm \tau(f)$. The existence of such a map τ is guaranteed by the following theorem.

Theorem 4.1. *Let $\mathcal{C}$ be the class of acyclic based chain complexes over R . There is an unique map*

$$\begin{aligned} \mathcal{C} &\rightarrow \tilde{K}_1(R) \\ C_* &\mapsto \tau(C_*), \end{aligned}$$

such that

- *If $f : C \rightarrow C'$ is a chain isomorphism, then $\tau(f) = \tau(C') - \tau(C)$;*
- *$\tau(C \oplus C') = \tau(C) + \tau(C')$;*

- $\tau(0 \rightarrow C_n \xrightarrow{f} C_{n-1} \rightarrow 0) = (-1)^n[f]$.

This can be related back to simple homotopy equivalences with the following definition.

Definition 7. Let M be an R -module and n be an integer. The elementary chain complex $E(M, n)$ is the complex defined by $E_n(M, n) = E_{n-1}(M, n) = M$, and $E_k(M, n) = 0$ for $k \neq n, n-1$. And with differential

$$\partial_n = 1_M : M \rightarrow M.$$

A simple chain complex is a direct sum of elementary chain complexes in which $M = R^{k_n}$.

As the name suggests, if $K \searrow^e L$, then $C(\tilde{K}, \tilde{L})$ is a elementary chain complex, and if $K \searrow L$ then $C(\tilde{K}, \tilde{L})$ is a simple chain complex.

Theorem 4.2. 1. If $f : C \rightarrow C'$ is an chain isomorphism between simple chain complexes, then $\tau(f) = 0$;

2. Let C be a finite, free, acyclic chain complex. There are simple chain complexes, D, E sand a chain isomorphism $f : D \rightarrow C \oplus E$.

Definition 8. Let C be a acyclic based chain complex. Define the torsion of C as

$$\tau(C) = \tau(f : D \rightarrow C \oplus E).$$

Corollary 4.1. If $f : D \rightarrow C \oplus E$ and $g : D' \rightarrow C \oplus E'$ are two such maps, then $\tau(f) = \tau(g)$, so $\tau(C)$ does not depend on the choice of D, E .

What if $f : C \rightarrow C'$ is not an isomorphism? What if f is only a weak equivalence?

Theorem 4.3. If $f : C \rightarrow C'$ is a weak equivalence between finite based chain complexes, then the cone of f , $C(f)$, is a finite acyclic based chain complex.

Definition 9. Let f be such as in the last theorem. Define

$$\tau(f) = \tau(C(f)).$$

5 Whitehead torsion of maps

If $f : K \rightarrow L$ is a homotopy equivalence between finite CW-complexes, then f is homotopic to a cellular homotopy equivalence $g : K \rightarrow L$, which in turn defines an weak equivalence of finite, based $\mathbb{Z}[\pi]$ -chain complexes

$$g_* : C_*(\tilde{K}) \rightarrow C_*(\tilde{L}),$$

and then we can define

$$\tau(f) = \tau(g_*) \in \text{Wh}(\pi_1(L)).$$

Theorem 5.1. 1. $\tau(f)$ is well-defined: it does not depend on the choice of orientation and lift used to fix generators of $C_*(\tilde{K}), C_*(\tilde{L})$, on the base point, on the identification of $\pi_1(L)$ with the group of covering transformations, or on the choice of cellular approximation g ;

2. f is a simple homotopy equivalence if and only if $\tau(f) = 0$.

6 H-cobordisms and the s-cobordism theorem

Definition 10. Let M, N be two n -dimensional manifolds. A h-cobordism is a quintuple (W, M_0, f_0, M_1, f_1) such that W is a $n+1$ -dimensional compact manifold which is a cobordism between M_0 and M_1 , i.e.

$$\partial W = M_0 \sqcup M_1,$$

and

$$f_0 : M_0 \rightarrow W, \quad f_1 : M_1 \rightarrow W,$$

are homotopy equivalences.

Theorem 6.1. *Let (W, M_0, f_0, M_1, f_1) be a h -cobordism, with W smooth and of dimension greater or equal to 6. If W is simply connected, then*

$$W \cong M_0 \times [0, 1].$$

W being simply connected is a big restriction; can we drop it?
 To a manifold we can give a CW-complex structure by a triangulation for instance, so the maps f_i are homotopy equivalences between two CW-complexes, and therefore we can calculate

$$\tau(W, M_0) = \tau(f_0) \in \text{Wh}(\pi_1(W)).$$

Theorem 6.2. *Let (W, M_0, f_0, M_1, f_1) be an h -cobordism, with W a of dimension greater or equal to 6. If W is smooth, then*

$$W \cong M_0 \times [0, 1],$$

if and only if $\tau(W, M_0) = 0$.

moreover, for every $x \in \text{Wh}(\pi_1(M_0))$, there is an h -cobordism W_x over M_0 such that $\tau(W_x, M_0) = x$. And if W, W' are two diffeomorphic h -cobordisms, then $\tau(W, M_0) = \tau(W', M_0)$, so τ gives us a bijection between diffeomorphism classes of h -cobordisms over M_0 and the group $\text{Wh}(\pi_1(M_0))$.

References